SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security COURSE CODE: CSA5117

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 **Total Marks:50**

Annexure A

CONCEPT MAPPING

Code of Conduct:

I K.S. DINESH (192321057) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Dinesh
Signature of the student:

Annexure A

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:
   a. Show how each attack leads to specific threats and risks
   b. Map how security goals help mitigate them Provide one real-world example for each link

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
   a. Show differences and similarities
      i. Map how each affects the Data structure and Security level
      ii. Analyze which is more vulnerable to attacks and why

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
   a. Show relationships between these concepts
   b. Analyze how each contributes to secure communication
   c. Include one real-world application for each

5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
   a. Show how each concept is mathematically connected
   b. Map their role in encryption/decryption
   c. Analyze which concept is most critical for public key cryptography and justify

RUBRICS FOR CALCULATION:

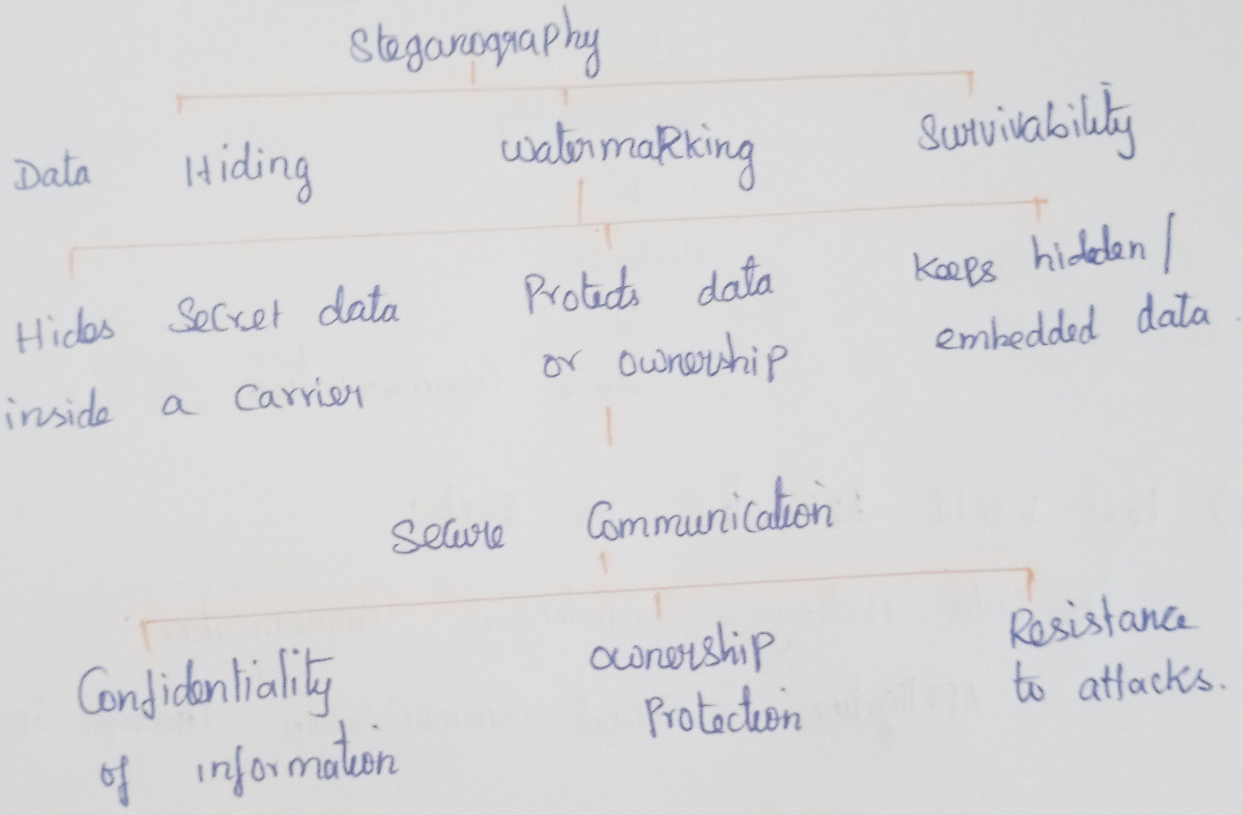
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

DINESH
192324185

1) Steganography Concepts:

Steganography is the technique of hiding secret information inside a normal-looking digital file such an image, audio, video or document.

Concept Map:



a) Relationship between the Concepts:

1) Data Hiding:
2) Water marking
3) Survivability
4) Connection

b) How they contribute to secure Communication:

Data Hiding

↓

Hides the existence of information

↓

Watermarking

↓

Provides authenticity

↓

Survivability

↓

Protects

↓

Secure Communication.

c) Real World Application for Each:

1) Data Hiding - Confidential Communication

Application : Secret Communication through images.

2) watermarking - Copyright Protection:

Application : Digital images and videos

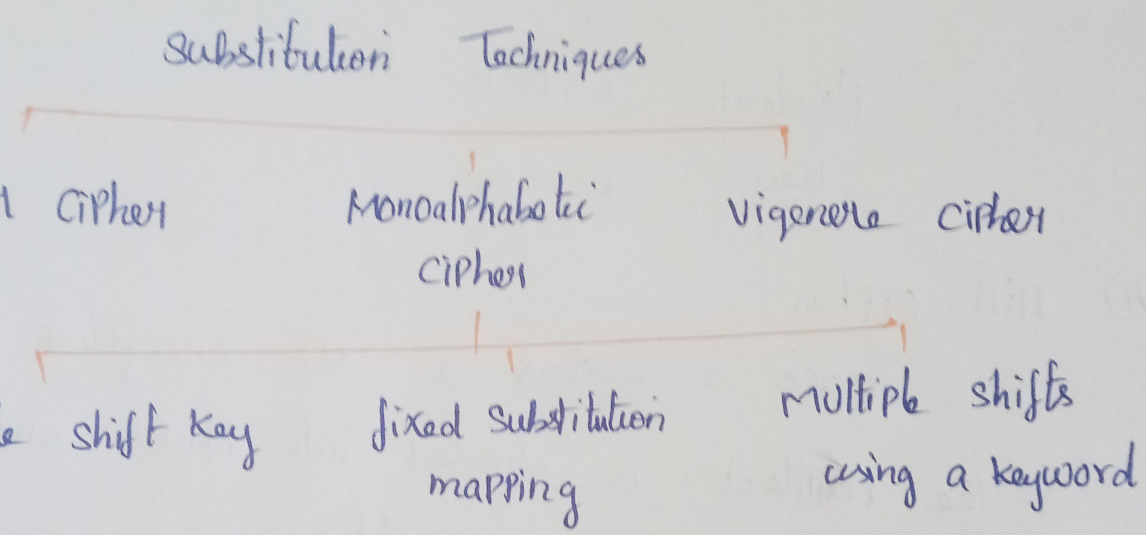
3) Survivability - Digital Media Protection:

Application : Video or image watermaking.

Substitution Techniques:

The four techniques are Caesar cipher, Monoalphabetic cipher, Vigenere Cipher and Hill Cipher. They can be compared based on key usage, Complexity.

Concept Map:



1) Caesar cipher:

uses a single numerical shift key.

Example with Key = 3

A → D, B → E, C → F

Complexity: very simple and easy to implement.

Security: very weak because there are only 25 Possible shifts and it can be broken by brute force.

2) Monoalphabetic cipher:

uses a fixed substitution alphabet

Ex: A → Q , B → W , C → E ...

Complexity : More Complex than Caesar Cipher.

3) Vigenere cipher:

uses a keyword to generated different shifts.

plaintext : ATTACK

Key : KEYKEY

4) Hill cipher:

uses a matrix as the encryption key.

The plaintext is divided into blocks and mathematical matrix multiplication is applied.

Security Ranking:

Caesar

↓

Monoalphabetics

↓

Vigenere

↓

Hill

Least Secure → Most Secure.